DEVELOPMENT OF A 3-DOF ROBOTIC ARM WITH IOT CONTROL SYSTEM

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Introduction

The following report is an overview of our project titled 'Development of a 3-DOF Robotic Arm with IoT Control System), where we discuss the development of the hardware model as well as a software simulation of the robot using original Denavit-Hartenberg parameters.

The objective of this project was to create a prototype of a functional robotic arm as used in various industrial applications, such as packing, filling and other processes in the manufacturing industry.

Hardware & Software Equipment Used

- ESP 32-S3 Microcontroller
- 3x SG-90 180^0 Servo Motor
- 1x SG-90 360^0 Servo Motor
- 5-V Power Supply
- Ice-cream sticks
- Rubber Flaps

- Jumper Wires
- Arduino IDE
- HTML-based app
- MATLAB

Procedure

1. Assembling the Robotic Arm Hardware

For the construction of the robotic arm, we fixed a wooden base so that the structure remains stable, for the structural part we used ice cream sticks and we used the rubber flaps to make the arm joints more flexible. For proportional control of the base, shoulder, and elbow joint, three miniature, 180° turn SG-90 servo motors were used while an additional 360° SG-90 motor was used to rotate the gripper horizontally for the open-close feature. These components were connected by jumper wires and were supplied by 5-V power supply.

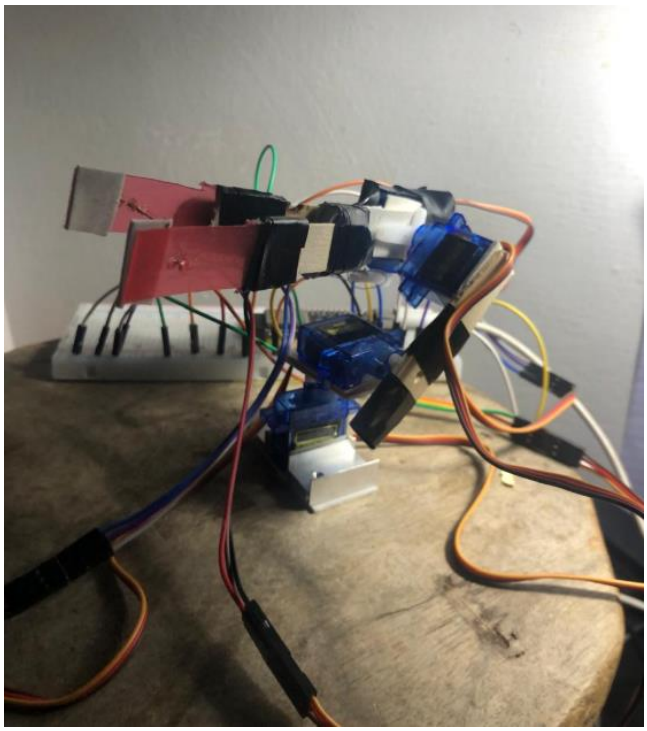


Figure 1 Arm with Elbow, Base and Shoulder Joints and End-Effector

2. Programming the ESP32-S3 Microcontroller

Here, we used the ESP32-S3 microcontroller employing Arduino IDE for controlling the servo motor through the signals. The microcontroller was programmed to decode inputs from an HTML application that allows manual control of the tilted position. Through sliders and button, we can control each of the joint and grippers of the robot and define the hard and soft home position.

To link the ESP32-S3 microcontroller with the IoT app, we created a wireless communication connection. The operation of servo motor movements is one of the crucial features that were tested and optimized to run smoothly. It, in turn, acted as a companion to the microcontroller, which interpreted commands in the shape of specific motor angles so as to orchestrate the arm's actions.

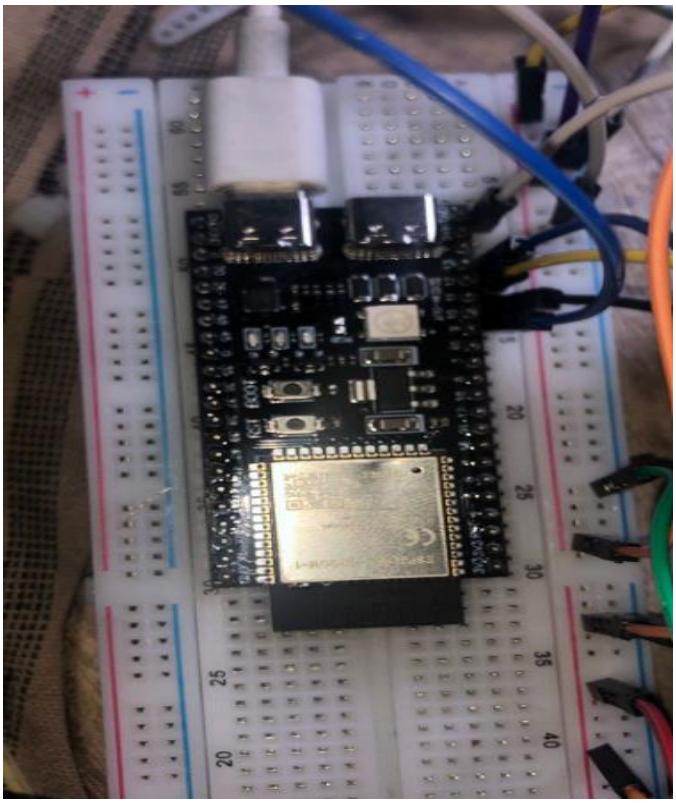


Figure 2 ESP 32 Microcontroller

3. Developing the IoT Control App

In addition, we created an HTML-based control application for the robotic arm. The servos were controlled via the app with sliders for the angles of the base, shoulder, and elbow

servos and buttons on the gripper. It was possible to store some particular positions of the arms, and set the hard and soft home positions concerning the frame, making it easy to control the arm from a distance.

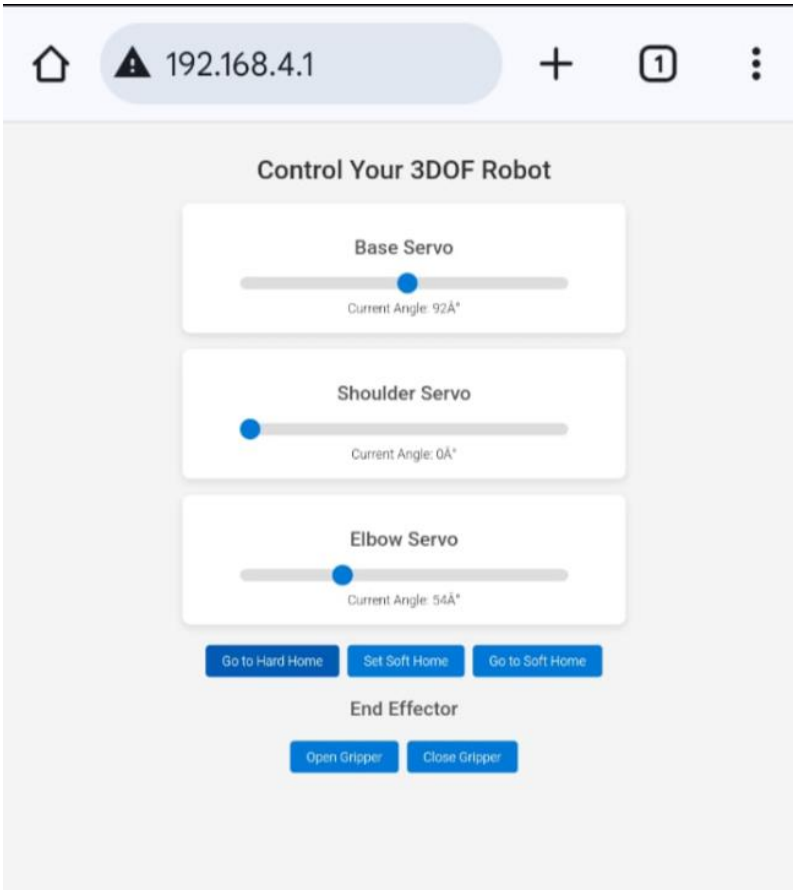
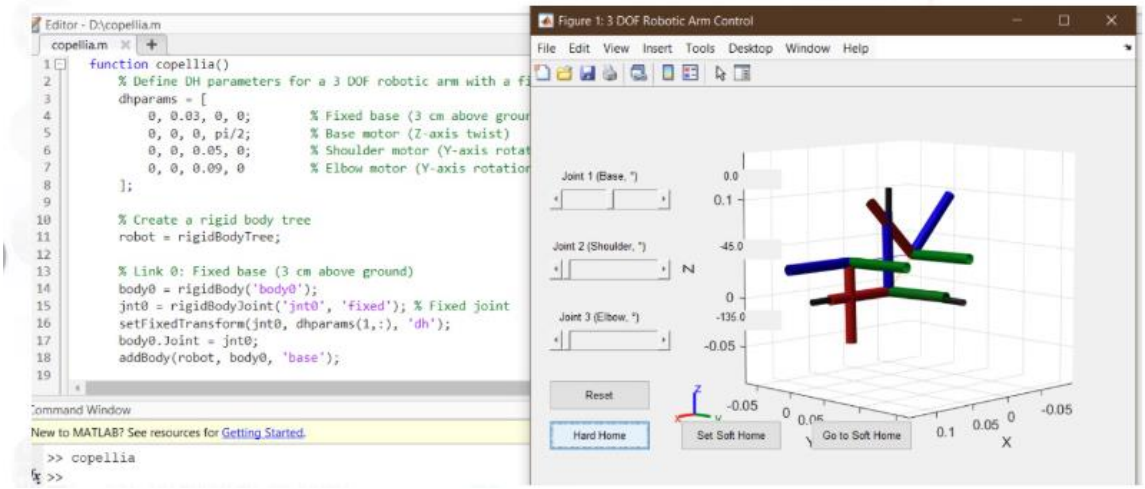


Figure 3 App UI

4. Simulating the Robotic Arm in MATLAB

MATLAB was used to prototype the robotic arm and its functions were modeled using a Rigid Body Tree model. Cartesian coordinate system was defined by Denavit-Hartenberg (DH) values, which are derived from the designed hardware, for describing the joint angles and link lengths. Through the use of simulation it was possible to observe the workspace, recognize singularities and visualize the reach and flexibility of the arm.

Note: MATLAB code attached in submission files



5. Validating and Optimizing the System

MATLAB was used to prototype the robotic arm and its functions were modeled using a Rigid Body Tree model. Cartesian coordinate system was defined by Denavit-Hartenberg (DH) values, which are derived from the designed hardware, for describing the joint angles and link lengths. Through the use of simulation it was possible to observe the workspace, recognize singularities and visualize the reach and flexibility of the arm.

Note: Demonstration Video attached in file.

Challenges

The challenges we faced and addressed during the development of the robotic arm were:

- Some structural choices such as ice-cream sticks were relatively ambitious, the possibility to change them was very limited, yet in some places these sticks could be reinforced, while the overall design gave enough support for the robotic arm.
- Low torque output as well as the jitter problem in SG-90 servo motors was partly overcome through the efficient programming of the sequence of movements and minimizing load on the motors for precise and orderly operations.
- Problems in the power supply were solved by regulating the use of motor loads and performing an analysis of the 5-V supply that eliminated fluctuations in the servo motor performance.
- Disconnection of wireless communication was addressed by minimizing the code of ESP32-S3 and repositioning the microcontroller to increase signal strength thus enabling proper app to hardware connectivity.
- Small differences in DH parameters were adjusted gradually through re-estimation of the measurements to make MATLAB simulation closer to the physical motion of the robotic arm.

- We initially restricted the area that the robotic arm could span through its planning by locating the base in the center of its operation field and fixing the length of its reach, which reduced the complexity of the control algorithm required.
- As mentioned earlier, the use of lightweight and modularity to build the arm provided the necessary flexibility to the structure of the arm which could be adjusted easily and that coupled with easy assembly also pointed towards the aspect of further enhancements to it.
- To communicate and control via the Web app developed in HTML was an astounding experience with the ESP32-S3, allowing instant and precise adjustments in response to servo motors.
- MATLAB's simulation capabilities revealed the arm's singularities and workspace, allowing us to fine-tune hardware configurations and enhance performance predictability.

The basic material and components' issues encouraged innovation in tackling more complex tasks, which in the end produced a stable and efficient robotic arm with IoT control at a low cost.

Lessons Learnt

A few interesting observations were made during the project, in correlation with future ideas that included:

- That idea is sensible because the IoT-based control system with the ESP32-S3 microcontroller and HTML app brought extensive compatibility for effective and precise control. Feedback arising from each test run facilitated continuous refinement of the system, to make the response time even better.
- Although there are severe disadvantages of using lightweight materials and components in such structures, conscious efforts were made to strengthen them and also assemble structures that would have enough stability during operation if the concept had to be further developed; there is thus enough evidence that support the use of low cost materials for prototyping.
- In the interaction with the app, servo motors displayed high levels of integration to the commands given. Suboptimal control sequences greatly influenced performance; adjusting their parameters eliminated erratic movements thus improving reliability.
- The limited available workspace for the arm meant that complex control algorithms were limited in use because operations were kept in a reasonable range but still provided functional flexibility.
- The possibility to design and create a mechanically constructive in sections granted future adaption, the plan already incorporated innovative ideas such as adding sensor's or upgraded material.

- Difficulties that arose during the project excluded the possibility of working only in the framework of a specific idea of the arm; on the contrary, they became the stimuli for the further evolution of the arm's design and its functions.

Conclusion

In conclusion, the development of the 3-DOF robotic arm with IoT control showcased a practical and cost-effective solution for prototyping industrial robotic applications. Despite material and hardware limitations, innovative problem-solving ensured stable operation and functional accuracy. The integration of ESP32-S3, HTML-based app control, and MATLAB simulation enabled precise motion control and insightful analysis of the arm's workspace and singularities. Challenges during the project fostered creative enhancements, highlighting the potential for scalability and future upgrades. This project successfully demonstrated the feasibility of lightweight, modular designs for robotics while providing valuable insights for advancing IoT-controlled systems in manufacturing and beyond.